

Applied Spatial Data Analysis

Chapter 3 - Spatial Lattice Data I

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ASDA: Spatial Lattice Data Analysis I

1. Introduction to lattice data and plotting
2. Spatial dependency
3. Defining spatial neighbours
4. Defining spatial weights matrices
5. Global Moran’s I
6. Geary’s C
7. Spatial correlogram
8. Local Moran’s I

Resources

For general spatial data handling in R using **simple features (sf)**.

Required reading: Chapter 10.

- [1] Anselin , Luc, (1999). Spatial Econometrics: Methods and Models. Luwer Academic Publishers.
- [2] Arbia , Giuseppe, (2014). A Primer for Spatial Econometrics : With Applications in R. Palgrave Macmillan UK. (available at UCT library online resource)
- [3] Elhorst , Paul. J. (2014). Spatial Econometrics : From Cross-Sectional Data to Spatial Panels. Springer Berlin
- [4] Gelfand, A.E., Diggle, P., Guttorp , P., Fuentes, M. and Fitzmaurice, G. (2010). Handbook of Spatial Statistics. Chapman & Hall/CRC. (available at UCT library online resource) Brilliant resource to have. Part 4 is dedicated to SPPs. *
- [6] Haining , Robert, (2004). Spatial Data Analysis, Theory and Practice. Cambridge University Press. *
- [7] <https://geodacenter.github.io/>

- Anselin, Luc (1999). *Spatial Econometrics: Methods and Models*. Kluwer Academic Publishers.
- Arbia, Giuseppe (2014). *A Primer for Spatial Econometrics: With Applications in R*. Palgrave Macmillan UK.
- Elhorst, Paul J. (2014). *Spatial Econometrics: From Cross-Sectional Data to Spatial Panels*. Springer.
- Gelfand, A. E., Diggle, P., Guttorp, P., Fuentes, M. & Fitzmaurice, G. (2010). *Handbook of Spatial Statistics*. Chapman & Hall/CRC.
- Haining, Robert (2004). *Spatial Data Analysis: Theory and Practice*. Cambridge University Press.
- GeoDa resources: <https://geodacenter.github.io/>

Example Datasets

Throughout these lectures we use the Boston house-price data, Columbus crime data, and EU regional data collected at NUTS 2 level.

- **Boston data:** classic Harrison and Rubinfeld (1978) housing data, with 506 census tracts and 23 variables.
- **Columbus data:** spatial crime data for Columbus.
- **EU2 data:** regional data for the second level of the Nomenclature of Territorial Units for Statistics (NUTS 2), widely used by Eurostat and other EU bodies.

A spatial data set includes the geometry of the spatial units, spatial coordinates or other spatial indexing information, and the attributes associated with each unit. These attributes may include economic, demographic, social, environmental, or other variables.

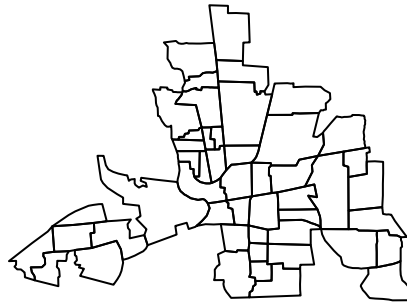
The three example data sets illustrate the same lattice-data idea: each observation belongs to a spatial unit, and each unit has both geometry and attributes. The geometry lets us construct neighbourhood relationships; the attributes are the variables we analyse.

A lattice data set is not just a data frame with coordinates. Its geometry carries the information needed to decide which areas are adjacent or close. In modern R workflows this geometry is usually stored in an `sf` object.

```
library(sf)
library(spData)

example(columbus)
```

```
colmbs> if (requireNamespace("sf", quietly = TRUE)) {
colmbs+   columbus <- sf::st_read(system.file("shapes/columbus.gpkg", package="spData")[1])
colmbs+   plot(sf::st_geometry(columbus))
colmbs+ }
Reading layer `columbus' from data source
  `C:\Users\01438475\AppData\Local\R\win-library\4.4\spData\shapes\columbus.gpkg'
  using driver `GPKG'
Simple feature collection with 49 features and 20 fields
Geometry type: POLYGON
Dimension:      XY
Bounding box:   xmin: 5.874907 ymin: 10.78863 xmax: 11.28742 ymax: 14.74245
Projected CRS:  Undefined Cartesian SRS with unknown unit
```



```
colmbs> if (requireNamespace("spdep", quietly = TRUE)) {
colmbs+   library(spdep)
colmbs+   col.gal.nb <- read.gal(system.file("weights/columbus.gal", package="spData")[1])
colmbs+ }
```

```
class(columbus)
```

```
[1] "sf"          "data.frame"
```

```
columbus
```

Simple feature collection with 49 features and 20 fields

Geometry type: POLYGON

Dimension: XY

Bounding box: xmin: 5.874907 ymin: 10.78863 xmax: 11.28742 ymax: 14.74245

Projected CRS: Undefined Cartesian SRS with unknown unit

First 10 features:

	AREA	PERIMETER	COLUMBUS_	COLUMBUS_I	POLYID	NEIG	HOVAL	INC	CRIME
1	0.309441	2.440629	2	5	1	5	80.467	19.531	15.725980
2	0.259329	2.236939	3	1	2	1	44.567	21.232	18.801754

```

3 0.192468 2.187547      4      6      3      6 26.350 15.956 30.626781
4 0.083841 1.427635      5      2      4      2 33.200  4.477 32.387760
5 0.488888 2.997133      6      7      5      7 23.225 11.252 50.731510
6 0.283079 2.335634      7      8      6      8 28.750 16.029 26.066658
7 0.257084 2.554577      8      4      7      4 75.000  8.438  0.178269
8 0.204954 2.139524      9      3      8      3 37.125 11.337 38.425858
9 0.500755 3.169707     10     18      9     18 52.600 17.586 30.515917
10 0.246689 2.087235     11     10     10     10 96.400 13.598 34.000835

```

```

      OPEN      PLUMB DISCBD      X      Y NSA NSB EW CP THOUS NEIGNO
1 2.850747 0.217155   5.03 38.80 44.07   1   1  1  0 1000   1005
2 5.296720 0.320581   4.27 35.62 42.38   1   1  0  0 1000   1001
3 4.534649 0.374404   3.89 39.82 41.18   1   1  1  0 1000   1006
4 0.394427 1.186944   3.70 36.50 40.52   1   1  0  0 1000   1002
5 0.405664 0.624596   2.83 40.01 38.00   1   1  1  0 1000   1007
6 0.563075 0.254130   3.78 43.75 39.28   1   1  1  0 1000   1008
7 0.000000 2.402402   2.74 33.36 38.41   1   1  0  0 1000   1004
8 3.483478 2.739726   2.89 36.71 38.71   1   1  0  0 1000   1003
9 0.527488 0.890736   3.17 43.44 35.92   1   1  1  0 1000   1018
10 1.548348 0.557724   4.33 47.61 36.42   1   1  1  0 1000   1010

```

geom

```

1 POLYGON ((8.624129 14.23698...
2 POLYGON ((8.25279 14.23694,...
3 POLYGON ((8.653305 14.00809...
4 POLYGON ((8.459499 13.82035...
5 POLYGON ((8.685274 13.63952...
6 POLYGON ((9.401384 13.5504,...
7 POLYGON ((8.037741 13.60752...
8 POLYGON ((8.247527 13.58651...
9 POLYGON ((9.333297 13.27242...
10 POLYGON ((10.08251 13.03377...

```

```
# The lecture data contain 49 Columbus neighbourhoods.
```

```
nrow(columbus)
```

```
[1] 49
```

```
names(columbus)
```

```

[1] "AREA"      "PERIMETER" "COLUMBUS_" "COLUMBUS_I" "POLYID"
[6] "NEIG"      "HOVAL"      "INC"        "CRIME"       "OPEN"

```

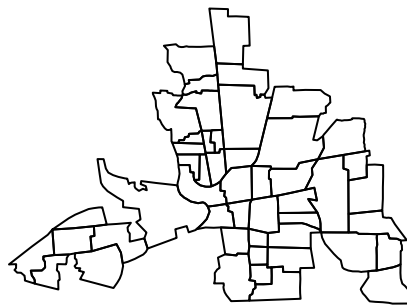
```
[11] "PLUMB"      "DISCBD"     "X"          "Y"          "NSA"
[16] "NSB"       "EW"         "CP"         "THOUS"      "NEIGNO"
[21] "geom"
```

Columbus Dataset

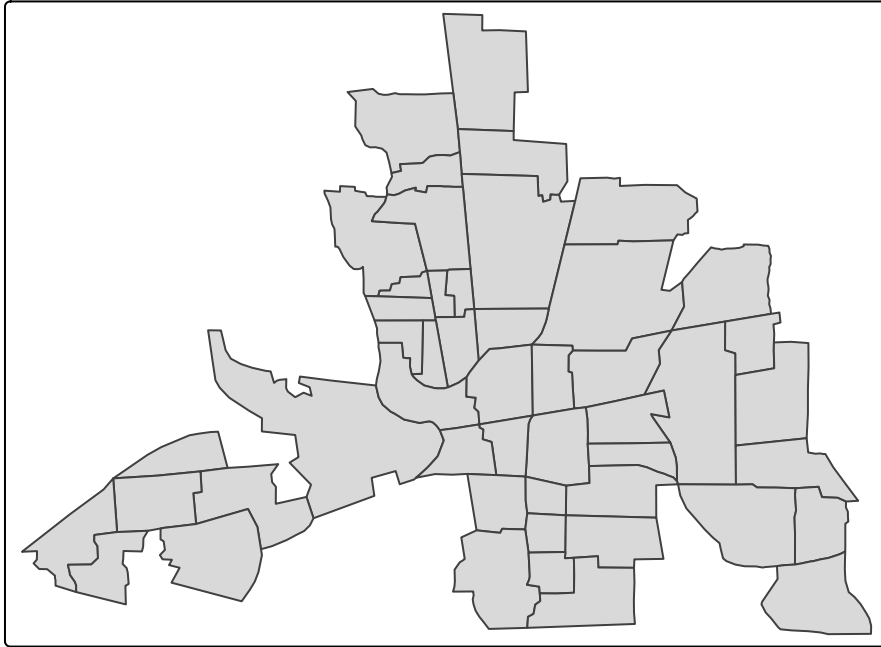
A major pleasure in working with spatial data is their visualisation. Maps are amongst the most compelling graphics.

The Columbus data provide a spatially referenced example used throughout the lecture.

```
plot(sf::st_geometry(columbus))
```

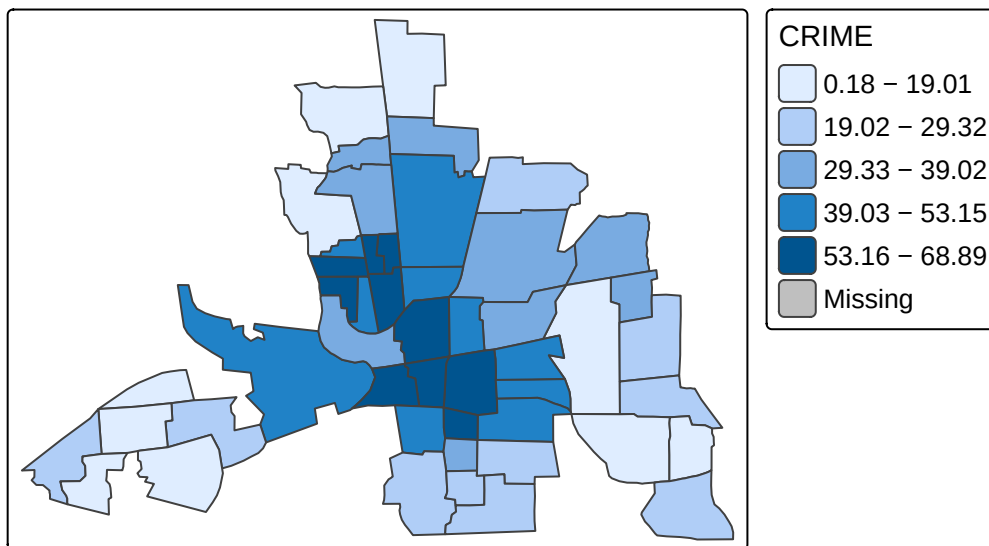


```
library(tmap)
tm_shape(columbus) + tm_polygons()
```



Plot of Crime Variable

```
tm_shape(columbus) + tm_polygons(style="quantile", col = "CRIME") +  
  tm_legend(outside = TRUE, text.size = .8)
```



EU Nuts2 Dataset

EU NUTS 2 level GDP per capita (Euros), 2009.



Analysing Spatial Lattice Data

Tobler's first law of geography:

“Everything is related to everything else, but close things are more related than things that are far apart.”

From this definition, we need to identify:

- what we mean by **spatial dependence**;
- how we measure spatial dependence;
- who the neighbours of each spatial unit are; and
- what the strength of the relationship is.

For each unit we therefore need a **list of neighbours** and a corresponding set of **spatial weights**.

Tobler's law is a motivation, not a statistical model. Spatial statistics makes it operational by defining *which* locations are close, *how strongly* they are connected, and whether neighbouring values are more similar than would be expected under a suitable null model.

Definition of Spatial Dependence

Spatial dependence in a collection of sample observations refers to the fact that an observation associated with location i depends on observations at locations j , where $i \neq j$.

Why does spatial dependence occur? Two main reasons are highlighted by LeSage (1998, pp. 4–5):

- measurement error related to the data-collection process for geographical units;
- the spatial dimension of the research problem may itself be important in explaining the process under study.

Spatial dependence may be substantive—because processes spill across borders—or partly induced by how data are measured and aggregated. This is one reason the choice of spatial unit and spatial weights needs substantive justification.

Spatial dependence reflects a situation where values observed in one areal unit, depend on the values of neighbouring observations at near-by areas.

Positive and Negative Spatial Dependence

Positive spatial dependence

Objects—points, areas, or lines—that are close in space tend to assume **similar values**. This produces spatial clustering of similar values.

Negative spatial dependence

Objects that are close in space tend to assume **dissimilar values**. This can produce an alternating or checkerboard-like arrangement.

Spatial dependence can therefore take the form of **spatial correlation**.

Ref: Giuseppe Arbia lecture notes, SEAI 2011.

Positive and negative dependence describe the *sign* of local association. They should not be confused with high and low values themselves: a low value surrounded by low values is still positive spatial association.

Exploring Spatial Dependence

The level of spatial dependence can be explored using graphical techniques and statistical measures:

- graphical representations of spatial dependence;
- Moran's I ;
- Geary's C .

Cliff and Ord (1972, 1975, 1981) found Moran's I to be a particularly useful measure in many settings.

Moran's I and Geary's C both combine attribute information with spatial relationships, but they emphasise different aspects. Moran's I works with cross-products of deviations from the mean; Geary's C works with squared pairwise differences.

Moran's I and Geary's C are very alike, and they both account for the attribute similarity and locational similarity. Moran's I looks at similarity, whereas Geary's C focuses on dissimilarity, similar to the notion of variogram as in geostatistics. Moran's I and Geary's C are interpreted in an opposite manner. Moran's I and Geary's C are very much linked to weights matrices. With correlograms and variograms, we will step away from weights matrices and look at distances. We still have attribute similarity however the locational similarity will be replaced with increasing distances between pairs of observations.

Global Moran's I Statistic

Global Moran's I combines **attribute similarity** and **locational similarity**.

For a variable y , let

$$z_i = y_i - \bar{y},$$

and let $W^s = (w_{ij}^s)$ denote the row standardized spatial weights matrix. The general form of Moran's I is

$$I = \frac{Z'W^sZ}{Z'Z}$$

where

$$\sum_{i,j=1}^n w_{ij}^s = n.$$

For a row-standardised weights matrix, the spatial lag W^sZ is the weighted mean of neighbouring standardised observations.

The spatial weights matrix is not a minor technical choice. It defines the spatial comparison being made. Changing W can change the value, interpretation, and significance of Moran's I .

Global Moran's I can be seen as a measure of spatial autocorrelation overall, as one measure, later on we will look at Local Moran's I which is more useful to look at specific locations and find hotspots etc. We can only use this statistic when 2 assumptions are satisfied: 1) Constant mean: We deal with that by mean standardizing our variable. Centring on the mean is equivalent to asserting that the correct model has a constant mean (zero mean since we mean standardized the variable into Z), and that any remaining patterning after centring is caused by the spatial relationships encoded in the spatial weights. The second assumption is constant variance, given in the denominator. There are some ways to deal with that as well. The value depends highly on the weights matrix chosen therefore we cannot compare different Moran's I statistics from different datasets or weights used. However we can convert the Moran's I statistic into a Z value which takes into account the differences in data and weights matrices used and then compare the Z values.

How to Construct Weights' Matrix

For geographic lattice data, there are two broad choices for W (Arbia, 2006, pp.37-38, Fischer et al., 2010, pp.66-68, LeSage, 1999, pp.11-13):

Boundary-based weights

- Rook
- Bishop
- Queen

Distance-based weights

- k -nearest-neighbour connectivity
- critical cut-off connectivity
- inverse-distance connectivity
- distance-decay functions
- combinations of these approaches

Boundary-based weights are often natural for administrative polygons. Distance-based weights can be preferable when interaction does not stop at administrative borders, or when the units are points/centroids rather than contiguous polygons.

Spatial Weights Based on Boundaries : These types of spatial weights define two observations connected if they share a common boundary. Contiguity based measures are more relevant for a polygon type data set or for observations that have a polygon attribute. There are three different ways to represent the connectivity based on the boundaries (Arbia , 2006, pp.37-38, Fischer et al., 2010, pp.66-68, LeSage, 1999, pp.11-13),such as: Spatial Weights Based on Distances: Binary or real valued spatial weights are based on the distances between the centroids of each observation. There are several approaches in defining the observations that are connected based on distances between them (Arbia , 2006, pp.37-38, Fischer et al., 2010, pp.66-68) K-nearest neighbour connectivity Critical cut-off based connectivity Inverse distance connectivity Distance decay function Combination of the several

Defining Spatial Neighbours: Boundary Based

Neighbourhood relationships can be defined in several ways (Arbia, 2005, p. 37).

For polygon data, a simple definition is **contiguity**: polygons i and j are neighbours when they share part of their boundary.

The three forms considered here are:

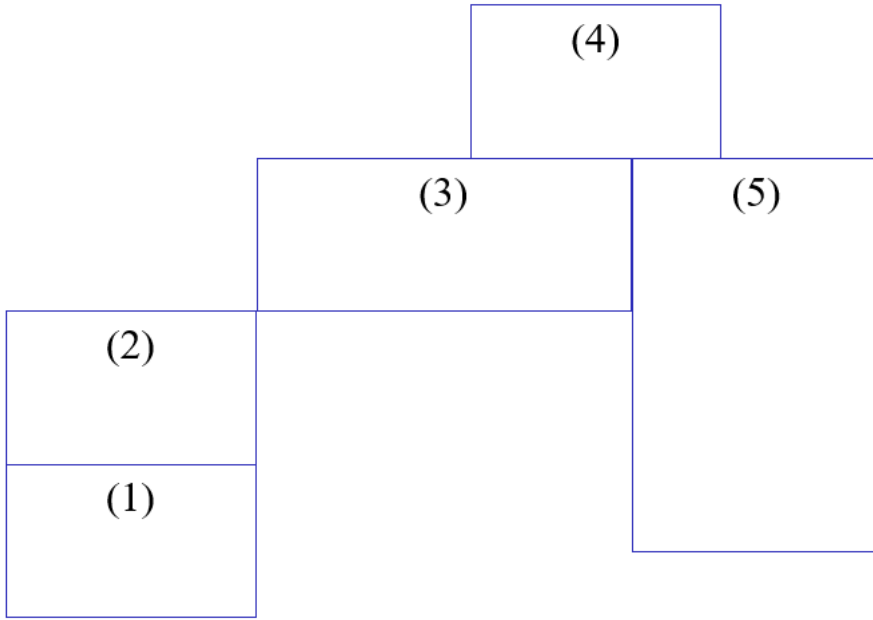
- **Rook**: common side;
- **Bishop**: common vertex only;
- **Queen**: common side or common vertex.

Robustness matters. If conclusions change dramatically when moving from rook to queen contiguity, or from one distance threshold to another, that sensitivity is itself an important empirical finding.

It needs to be stressed here that the result of any econometric analysis will be dependent on the specific topology (and, hence, of the neighbouring structure) chosen for the random field. Consequently it is always wise to test the robustness of the results obtained by adopting several definitions of neighbourhood.

Spatial Weights Based on Boundaries

A hypothetical five-region configuration is used to construct and interpret a binary contiguity matrix W .



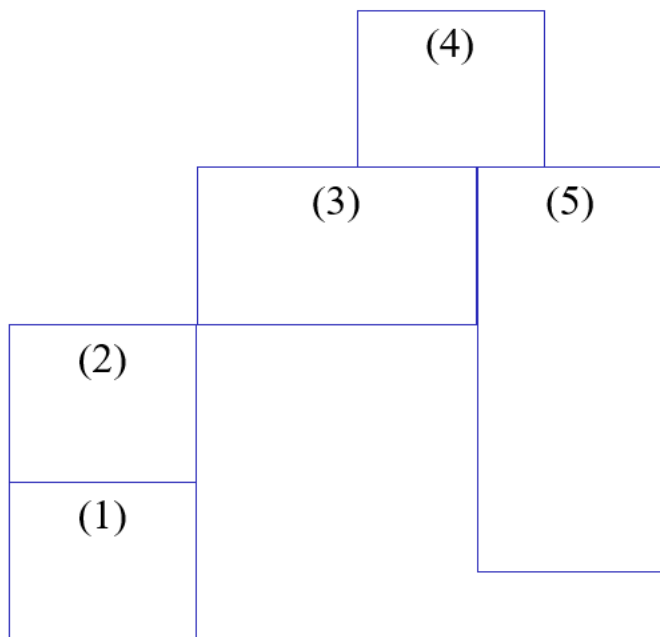
Le Sage page: 9 Figure 1.2 shows a hypothetical example of five regions as they would appear on a map. We wish to construct a 5 by 5 binary matrix W containing 25 elements taking values of 0 or 1 that captures the notion of “connectiveness” between the five entities depicted in the map configuration. We record in each row of the matrix W a set of contiguity relations associated with one of the five regions. For example the matrix element in row 1, column 2 would record the presence (represented by a 1) or absence (denoted by 0) of a contiguity relationship between regions 1 and 2. As another example, the row 3, column 4 element would reflect the presence or absence of contiguity between regions 3 and 4. Of course, a matrix constructed in such fashion must be symmetric | if regions 3 and 4 are contiguous, so are regions 4 and 3.

Spatial Weights Based on Boundaries: Rook Contiguity

Rook contiguity: define

$$w_{ij} = \begin{cases} 1, & \text{if regions } i \text{ and } j \text{ share a common side,} \\ 0, & \text{otherwise.} \end{cases}$$

The diagonal is normally set to zero, $w_{ii} = 0$, because a region is not treated as its own neighbour.



- **Rook contiguity:** Define $w_{ij} = 1$ for regions that share a **common side** with the region of interest.

- $$W = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

- $$W^s = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1/2 & 1/2 & 0 \end{bmatrix}$$

Rook contiguity is stricter than queen contiguity because touching at a corner is not enough. For grid cells this corresponds to the four orthogonal directions.

```
library(sf)
library(spdep)

# For a polygon sf object called x:
rook_nb <- poly2nb(columbus, queen = FALSE)
rook_w <- nb2listw(rook_nb, style = "W")
head(rook_nb)
```

```
[[1]]
[1] 2 3
```

```
[[2]]
[1] 1 3 4
```

```
[[3]]
[1] 1 2 4 5
```

```
[[4]]
[1] 2 3 5 8
```

```
[[5]]  
[1] 3 4 6 8 9 11 15
```

```
[[6]]  
[1] 5 9
```

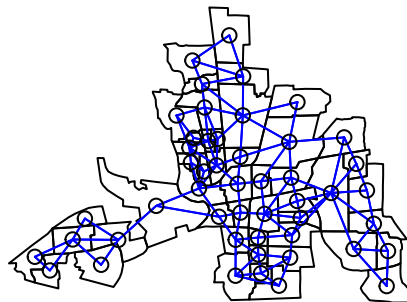
```
rook_w$neighbours[[1]]
```

```
[1] 2 3
```

Obtain the Centroids

```
coords <- st_coordinates(st_centroid(st_geometry(columbus)))
```

```
plot(st_geometry(columbus))  
plot(rook_nb, coords, add=TRUE, col="blue")
```

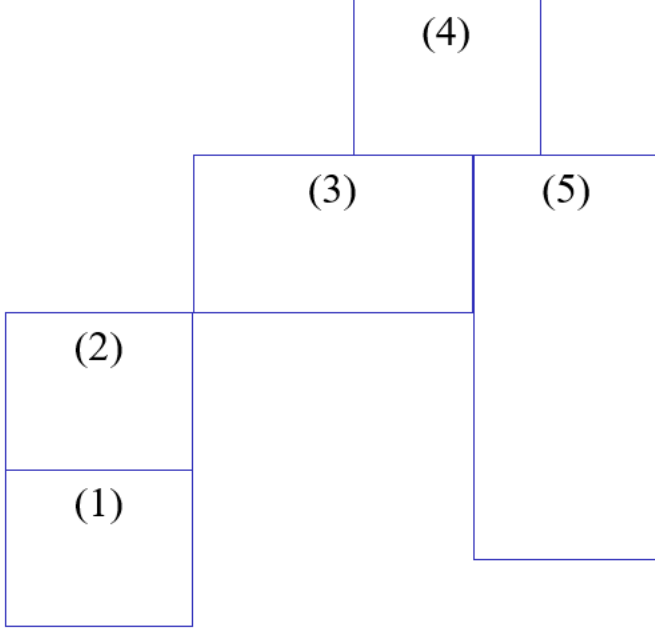


A transformation often used in applied work is to convert the matrix W to have row-sums of unity. This is referred to as a “standardized first-order” contiguity matrix, which we denote as W_s

Spatial Weights Based on Boundaries: Bishop Contiguity

Bishop contiguity: define

$$w_{ij} = \begin{cases} 1, & \text{if regions } i \text{ and } j \text{ share a common vertex only,} \\ 0, & \text{otherwise.} \end{cases}$$



- **Bishop contiguity:** Define $w_{ij} = 1$ for regions that share a **common vertex** with the region of interest.

- $W = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

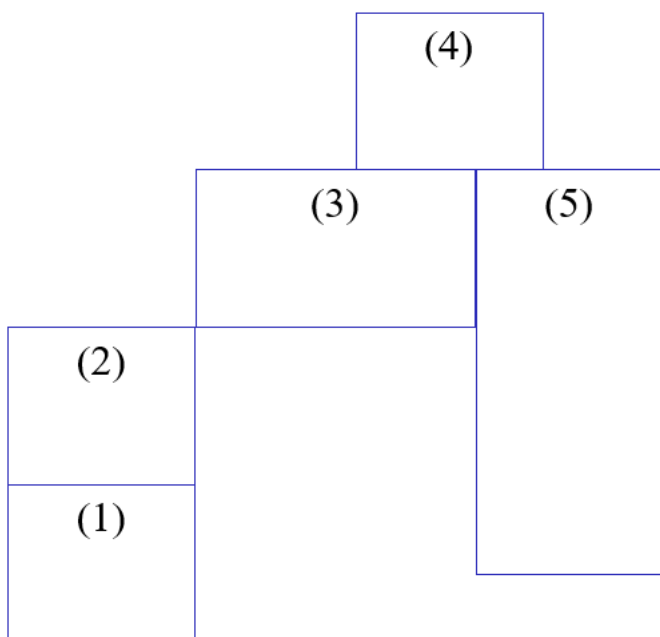
- $W^s = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Bishop contiguity is rarely used on its own in applied lattice analysis, but it is useful pedagogically because it isolates vertex-only contact.

Spatial Weights Based on Boundaries: Queen Contiguity

Queen contiguity: define

$$w_{ij} = \begin{cases} 1, & \text{if regions } i \text{ and } j \text{ share a common side or vertex,} \\ 0, & \text{otherwise.} \end{cases}$$



- **Queen contiguity:** Define $w_{ij} = 1$ for regions that share a **common side or vertex** with the region of interest.

- $$W = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

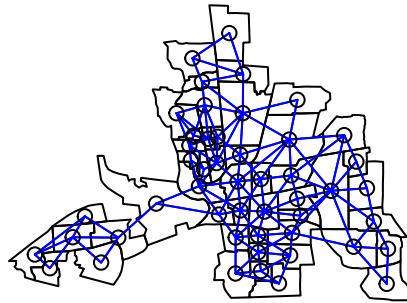
- $$W^s = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1/3 & 0 & 1/3 & 1/3 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1/2 & 1/2 & 0 \end{bmatrix}$$

Queen contiguity is the union of rook and bishop relationships. On a regular grid it corresponds to up to eight surrounding cells.

```
library(spdep)

queen_nb <- poly2nb(columbus, queen = TRUE)
queen_w <- nb2listw(queen_nb, style = "W")

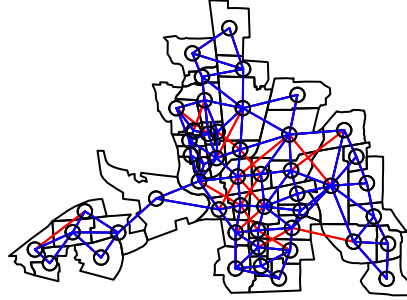
plot(st_geometry(columbus))
plot(queen_nb, coords, add=TRUE, col="blue")
```



Believe it or not, there are even more ways one could proceed.

Difference between Queen and Rook

```
plot(st_geometry(columbus))  
plot(queen_nb, coords, add=TRUE, col="red")  
plot(rook_nb, coords, add=TRUE, col="blue")
```



Defining Spatial Neighbours – Distance Based

Neighbourhood relationships may also be defined by distance.

***k*-nearest-neighbour neighbourhood**

For each spatial unit i , choose the k observations whose representative locations are closest to i . **(k)-nearest-neighbour neighbourhood**

For the first-nearest-neighbour case,

$$w_{ij} = \begin{cases} 1, & \text{if } d_{ij} = \min_{k \neq i} d_{ik}, \\ 0, & \text{otherwise.} \end{cases}$$

Critical cut-off neighbourhood

Two units (i) and (j) are neighbours if

$$w_{ij} = \begin{cases} 1, & \text{if } d_{ij} < d^*, \\ 0, & \text{otherwise.} \end{cases}$$

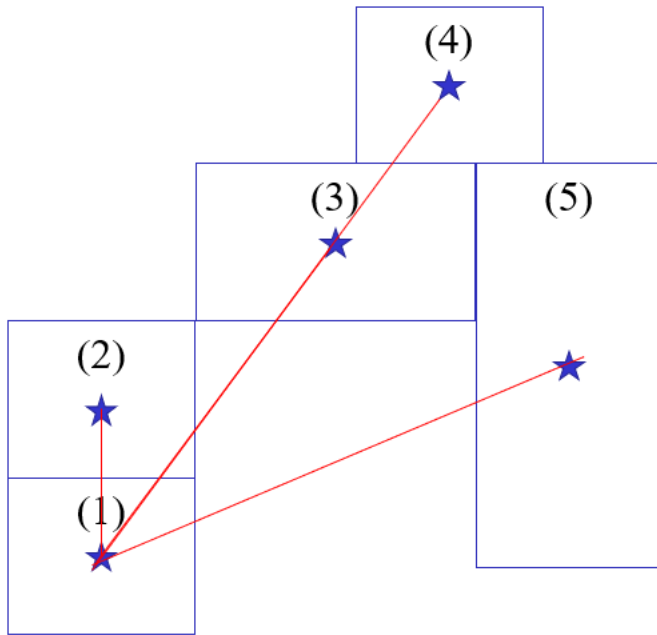
where d_{ij} is the adopted distance measure and d^* is a chosen critical cut-off.

For k -nearest neighbours every observation can be given exactly k outgoing neighbour links, which avoids isolates. A fixed distance threshold instead preserves a common physical scale but may create very different neighbour counts across dense and sparse areas.

where distances are calculated from information on latitude and longitude of the centroid locations. Examples include straight-line distances, great circle distances, travel distances or times, and other spatial separation measures. Straight-line distances determine the shortest distance between any two point locations in a flat plane, treating longitude and latitude of a location as if they were equivalent to plane coordinates. In contrast, great circle distances determine distances between any two points on a spherical surface such as the earth as the length of the arc of the great circle between them (see Longley et al. 2001, pp. 86–92, for more details). In many applications the simple measures—straight-line distances and great circle distances—are not sufficiently accurate estimates of actual travel distances, and one is forced to resort to summing the actual lengths of travel routes, using a GISystem . This normally means summing the length of links in a network representation of a transportation system.

Spatial Weights Based on Distances: 1st Nearest Neighbour – Obs (1)

For observation (1), calculate pairwise centroid distances and identify the closest observation. The first-nearest-neighbour link is then entered in the neighbour structure.



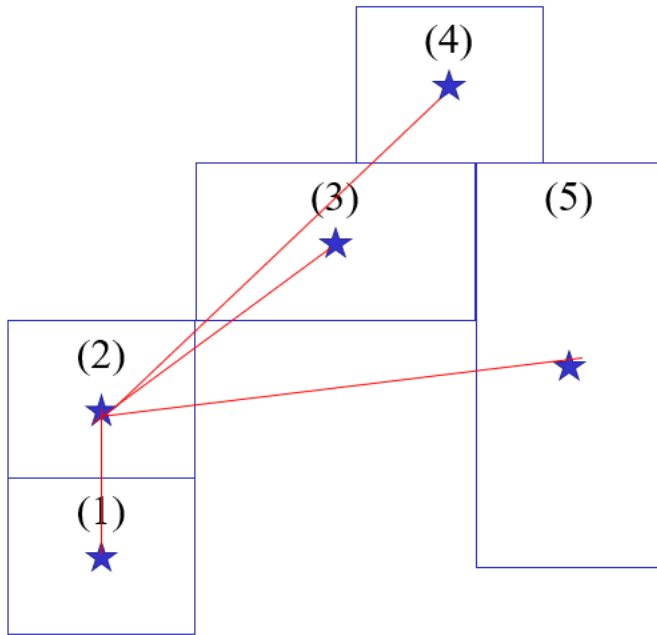
- Define $w_{ij} = 1$ for regions that are the closest in order. The pairwise distances are calculated (using the centroids).

- $d_{ij} = \begin{bmatrix} 0 & 2.5 & 4.2 & 5.8 & 4.5 \\ 2.5 & 0 & \square & \square & \square \\ 4.2 & \square & 0 & \square & \square \\ 5.8 & \square & \square & 0 & \square \\ 4.5 & \square & \square & \square & 0 \end{bmatrix}$

- $W = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square \end{bmatrix}$

Spatial Weights Based on Distances: 1st Nearest Neighbour – Obs (2)

For observation (2), calculate pairwise centroid distances and identify the closest observation. The first-nearest-neighbour link is then entered in the neighbour structure.



- Define $w_{ij} = 1$ for regions that are the closest in order. The pairwise distances are calculated (using the centroids).

- $d_{ij} =$

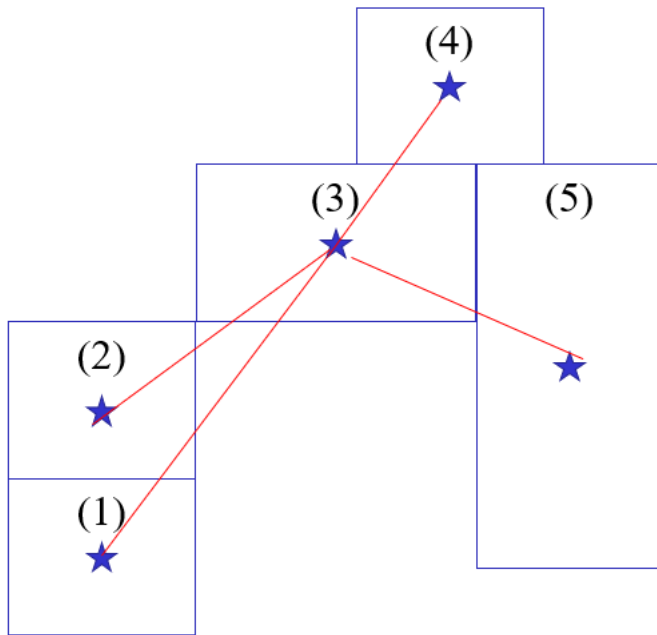
$$\begin{bmatrix} 0 & 2.5 & 4.2 & 5.8 & 4.5 \\ 2.5 & 0 & 3.8 & 5.7 & 5.1 \\ 4.2 & 3.8 & 0 & \square & \square \\ 5.8 & 5.7 & \square & 0 & \square \\ 4.5 & 5.1 & \square & \square & 0 \end{bmatrix}$$

- $W =$

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square \end{bmatrix}$$

Spatial Weights Based on Distances: 1st Nearest Neighbour – Obs (3)

For observation (3), calculate pairwise centroid distances and identify the closest observation. The first-nearest-neighbour link is then entered in the neighbour structure.



- Define $w_{ij} = 1$ for regions that are the closest in order. The pairwise distances are calculated (using the centroids).

- $d_{ij} =$

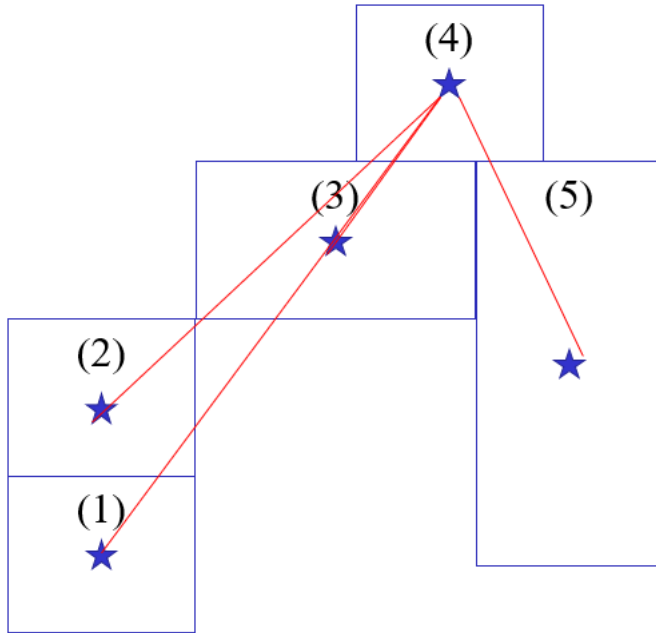
$$\begin{bmatrix} 0 & 2.5 & 4.2 & 5.8 & 4.5 \\ 2.5 & 0 & 3.8 & 5.7 & 5.1 \\ 4.2 & 3.8 & 0 & 2.6 & 3.6 \\ 5.8 & 5.7 & 2.6 & 0 & \square \\ 4.5 & 5.1 & 3.6 & \square & 0 \end{bmatrix}$$

- $W =$

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ \square & \square & \square & \square & \square \\ \square & \square & \square & \square & \square \end{bmatrix}$$

Spatial Weights Based on Distances: 1st Nearest Neighbour – Obs (4)

For observation (4), calculate pairwise centroid distances and identify the closest observation. The first-nearest-neighbour link is then entered in the neighbour structure.



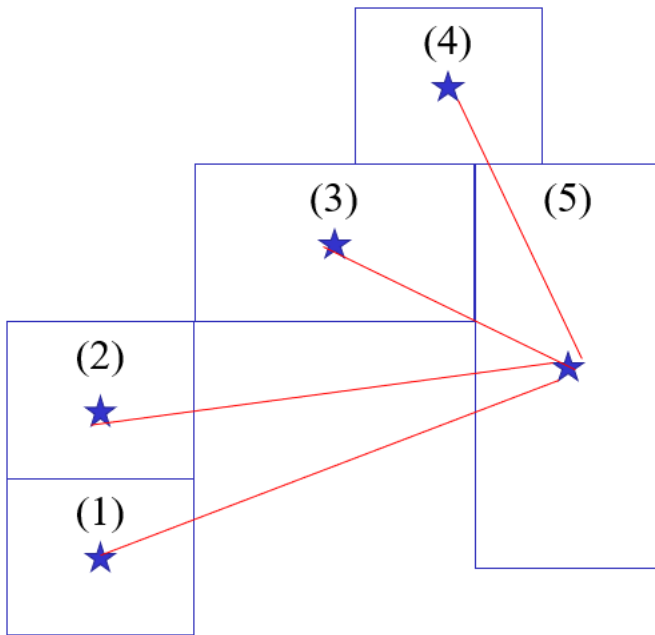
- Define $w_{ij} = 1$ for regions that are the closest in order. The pairwise distances are calculated (using the centroids).

- $d_{ij} = \begin{bmatrix} 0 & 2.5 & 4.2 & 5.8 & 4.5 \\ 2.5 & 0 & 3.8 & 5.7 & 5.1 \\ 4.2 & 3.8 & 0 & 2.6 & 3.6 \\ 5.8 & 5.7 & 2.6 & 0 & 4.3 \\ 4.5 & 5.1 & 3.6 & 4.3 & 0 \end{bmatrix}$

- $W = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ \square & \square & \square & \square & \square \end{bmatrix}$

Spatial Weights Based on Distances: 1st Nearest Neighbour – Obs (5)

For observation (5), calculate pairwise centroid distances and identify the closest observation. The first-nearest-neighbour link is then entered in the neighbour structure.



- Define $w_{ij} = 1$ for regions that are the closest in order. The pairwise distances are calculated (using the centroids).

- $d_{ij} =$

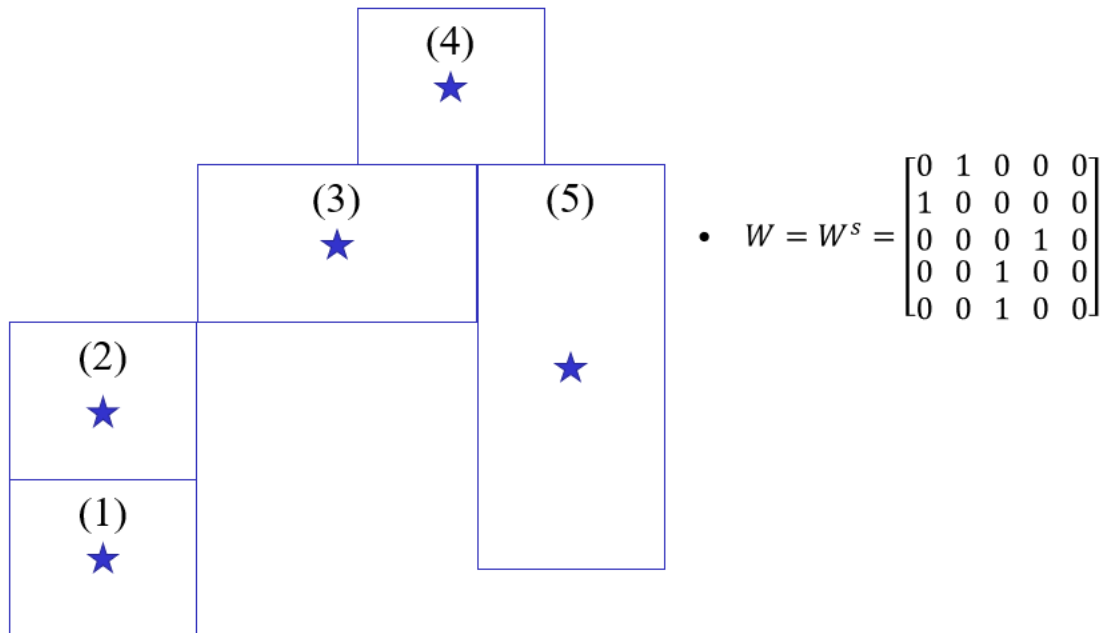
$$\begin{bmatrix} 0 & 2.5 & 4.2 & 5.8 & 4.5 \\ 2.5 & 0 & 3.8 & 5.7 & 5.1 \\ 4.2 & 3.8 & 0 & 2.6 & 3.6 \\ 5.8 & 5.7 & 2.6 & 0 & 4.3 \\ 4.5 & 5.1 & 3.6 & 4.3 & 0 \end{bmatrix}$$

- $W =$

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

Spatial Weights Based on Distances: 1st Nearest Neighbour

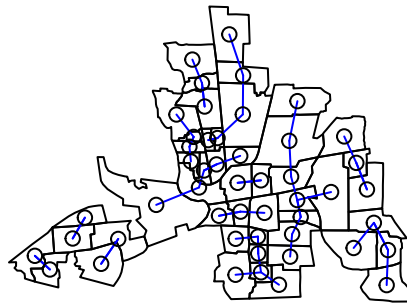
Combining the first-nearest-neighbour relationships for all five observations gives the complete first-nearest-neighbour graph.



Columbus Dataset: 1st Nearest Neighbours

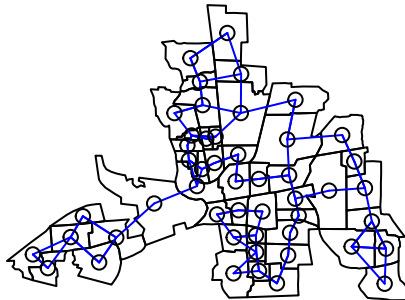
```
library(sf)
library(spdep)

whoisthefirstnear=knearneigh(coords,k=1,longlat=TRUE)
knn1columbus=knn2nb(whoisthefirstnear)
plot(st_geometry(columbus))
plot(knn1columbus, coords, add=TRUE, col="blue")
```



Columbus Dataset: 2 Nearest Neighbours

```
whois2near=knearneigh(coords,k=2,longlat=TRUE)
knn2columbus=knn2nb(whois2near)
plot(st_geometry(columbus))
plot(knn2columbus, coords, add=TRUE, col="blue")
```



Critical cut-off neighbourhood

To define neighbours within a fixed distance, we need a **threshold distance**.

One way of selecting a threshold is:

1. calculate the centroid of each polygon;
2. calculate the distance from each centroid to its first nearest neighbour;

```
distBetwNeigh1=nbdists(knn1columbus,coords,longlat=TRUE)
distBetwNeigh1=unlist(distBetwNeigh1)
distBetwNeigh1
```

```
[1] 63.98112 37.46759 57.90275 34.99180 50.68549 60.02289 41.50970 34.99180
[9] 53.16288 42.98225 13.85976 13.85976 15.87821 15.87821 36.74052 20.37838
[17] 42.32020 20.37838 21.89658 50.76316 67.50447 35.75701 42.32020 26.18731
[25] 34.14447 34.14447 25.62287 34.90927 32.08072 32.08072 35.72122 45.38893
[33] 25.62287 44.54357 28.40851 35.72122 32.67643 25.12110 29.91036 47.60799
[41] 45.38893 44.54357 25.12110 33.47541 37.48757 29.91036 51.94091 28.12354
[49] 32.04551
```

3. take the maximum of these first-neighbour distances.

For the Columbus example, the maximum first-nearest-neighbour distance is approximately

```
all.linkedTresh=max(distBetwNeigh1)
all.linkedTresh
```

```
[1] 67.50447
```

```
dnbTresh1=dnearneigh(coords,0,68,longlat=TRUE)
summary(dnbTresh1)
```

Neighbour list object:

Number of regions: 49

Number of nonzero links: 252

Percentage nonzero weights: 10.49563

Average number of links: 5.142857

2 disjoint connected subgraphs

Link number distribution:

```
1 2 3 4 5 6 7 8 9 10 11
4 8 6 2 5 8 6 2 6 1 1
```

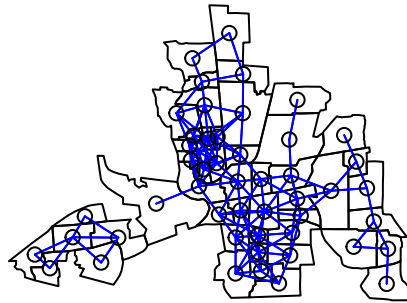
4 least connected regions:

6 10 21 47 with 1 link

1 most connected region:

28 with 11 links

```
plot(st_geometry(columbus))
plot(dnbTresh1, coords, add=TRUE, col="blue")
```



If you examine this neighbourhood list object a bit more closely you will see that, Polygon1 is neighbours with Poly2 and Poly3; Polygon2 is neighbours with Poly1 and Poly4; Polygon3 is neighbours with Poly1, Poly4 and Poly5 and so on:

```
dnbTresh1[1:3]
```

```
[[1]]
[1] 2 3
```

```
[[2]]
[1] 1 4
```

```
[[3]]
[1] 1 4 5
```

In calculation of the weights that will be used to create the spatially lagged variables, only the neighbours values are going to be used. For example for the 1st Polygon, the values of neighbouring Poly2 and Poly3 will be used. Before doing this the nb values need to be standardized, so that the row

Neighbours to Weights Matrix

Here we will work with the Critical Cut-off nearest neighbours

```
dnbTresh1.listw=nb2listw(dnbTresh1,style="W",zero.policy=FALSE)
class(dnbTresh1.listw)
```

```
[1] "listw" "nb"
```

```
dnbTresh1.listw$weights[1:5]
```

```
[[1]]
```

```
[1] 0.5 0.5
```

```
[[2]]
```

```
[1] 0.5 0.5
```

```
[[3]]
```

```
[1] 0.3333333 0.3333333 0.3333333
```

```
[[4]]
```

```
[1] 0.25 0.25 0.25 0.25
```

```
[[5]]
```

```
[1] 0.2 0.2 0.2 0.2 0.2
```

In matrix format rather than a list:

```
listw2mat(dnbTresh1.listw)[1:5,1:5]
```

	1	2	3	4	5
1	0.0000000	0.50	0.50	0.0000000	0.0000000
2	0.5000000	0.00	0.00	0.5000000	0.0000000
3	0.3333333	0.00	0.00	0.3333333	0.3333333
4	0.0000000	0.25	0.25	0.0000000	0.0000000
5	0.0000000	0.00	0.20	0.0000000	0.0000000

Choosing a threshold of 68 therefore guarantees that every centroid has at least one neighbour. As you can see row sum of the W^s is always 1, we row-standardized the W matrix.

$$W^s = \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 & \dots & 0 & 0 \\ 0.5 & 0 & 0 & 0.5 & 0 & \dots & 0 & 0 \\ 0.333 & 0 & 0 & 0.333 & 0.333 & \dots & 0 & 0 \\ 0 & 0.25 & 0.25 & 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \dots & 0 & 0 \end{bmatrix}_{(49 \times 49)}$$

Choosing the maximum first-nearest-neighbour distance as the threshold is a practical way to prevent isolates. It does **not** mean that this threshold is substantively optimal; it is a connectivity-based choice that should still be checked against the research problem.

Row standardisation makes every row sum to one, so each area's neighbours collectively receive the same total weight. A consequence is that a region with two neighbours gives each neighbour more weight than a region with ten neighbours, all else equal.

```
W_mat <- listw2mat(dnbTresh1.listw)

rowSums(W_mat)
```

```
1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26
1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49
1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1
```

Exploring Spatial Dependence

The level of spatial dependence (spatial autocorrelation) can be explored with:

- Moran's I ;
- Geary's C ;
- graphical representations such as the Moran scatter plot, correlogram, and variogram.

A spatial autocorrelation test asks whether the observed arrangement of values is more spatially structured than expected under a null model. Analytical tests and permutation tests use different routes to construct that reference distribution.

Spatial autocorrelation tests are decision rules based on statistics such as Moran's I and Geary's C to assess the extent to which the observed spatial arrangement of data values departs from the null hypothesis that space does not matter. This hypothesis implies that near-by areas do not

affect one another such that there is independence and spatial randomness. In contrast, under the alternative hypothesis of spatial autocorrelation (spatial association, spatial dependence), the interest renders on cases where large values are surrounded by other large values in near-by areas, or small values are surrounded by large values and vice versa. The former is referred to as positive spatial autocorrelation, and the latter as negative spatial autocorrelation. Positive spatial autocorrelation implies a spatial clustering of similar values, while negative spatial autocorrelation implies a checkerboard pattern of values

Some notation

To simplify notation:

- n : number of spatial areas;
- i, j : two areal units;
- y_i : value of the variable of interest at area i ;
- $z_i = y_i - \bar{y}$: mean-centred value at area i ;
- w_{ij} : measure of spatial connection between i and j ;
- $W = (w_{ij})$: spatial weights matrix;
- $W^{(s)}$: row-standardised spatial weights matrix.

Global Measures of Spatial Autocorrelation: Moran's I Statistic

Moran's I is a **global measure**: it provides one summary of spatial autocorrelation for the entire spatial pattern.

$$I = \frac{Z'W^sZ}{Z'Z}$$

With row-standardised W^s , this is closely related to the cross-product between the centred variable $Z' = Y - \bar{Y}$ and its spatial lag W^sZ .

The numerator of Moran's I is large and positive when connected areas tend to have deviations from the mean with the same sign. It becomes negative when neighbours tend to fall on opposite sides of the mean.

Spatial autocorrelation is considered to be present when the spatial autocorrelation statistic computed for a particular pattern takes on a larger value, compared to what would be expected under the null hypothesis of no spatial association.

Moran's I Expected Value and Variance

Under the randomisation null hypothesis, the expected value of Moran's I is

$$E(I) = -\frac{1}{n-1}.$$

Thus, under the null, Moran's I is not centred exactly at zero for a finite sample. As n increases,

$$E(I) \longrightarrow 0.$$

A standardised test statistic can be constructed as

$$Z_I = \frac{I - E(I)}{\sqrt{\text{Var}(I)}}.$$

The variance depends on the spatial weights and on the assumptions used for the reference distribution.

Ref: Fischer & Wang (2011), p. 25.

The variance of Moran's I

$$\text{Var}(I) = \frac{NS_4 - S_3S_5}{(N-1)(N-2)(N-3)W^2} - [E(I)]^2$$

where

$$S_1 = \frac{1}{2} \sum_i \sum_j (w_{ij} - w_{ji})^2$$

$$S_2 = \sum_i \left(\sum_j w_{ij} - \sum_j w_{ji} \right)^2$$

$$S_3 = \frac{N^{-1} \sum_i (x_i - \bar{x})^4}{\left[N^{-1} \sum_i (x_i - \bar{x})^2 \right]^2}$$

and

$$S_4 = (N^2 - 3N + 3)S_1 - NS_2 + 3W^2$$

$$S_5 = (N^2 - N)S_1 - 2NS_2 + 6W^2$$

Because $E(I) = -1/(n-1)$, comparing an observed I directly with zero is only an approximation. Formal inference compares the observed statistic with its expected value and variability under the selected null model.

Under spatial random process, the expected value of Moran's I is given as follows. As you can see it is not centred around 0, it is a bit off to the negative side. And it depends on n , sample size. As n becomes larger and larger, Moran's I gets closer to zero. The key question is the statistic that we obtained from the data, does it come from a spatial random process or not? In order to do the test we need the variance. As I mentioned earlier $E(I)/\text{Var}(I)$ is comparable since the variance takes into account the different W values.

Comparing Moran's I

Two broad approaches are used to assess whether an observed Moran's I departs significantly from the null hypothesis of no spatial autocorrelation:

1. random permutation;
2. an approximate sampling distribution for I .

Ref: [lecture comparison example and linked video, around minute 22:25](#).

In principle, there are two main approaches to testing observed I -values for significant departure from the hypothesis of zero spatial autocorrelation. 1) Random permutation test 2) approximate sampling distribution of the I values.

Moran's I Statistics

- A Moran's I value **above its expected value** indicates positive spatial autocorrelation.
- A Moran's I value **below its expected value** indicates negative spatial autocorrelation.
- Moran's I is a **global** measure: it does not by itself identify where local clusters or outliers occur.

The magnitude of a raw Moran's I should not be compared casually across different data sets or different weights matrices. For inference, use the reference distribution appropriate to the selected W and null hypothesis.

Calculation of Moran's I for the Columbus Dataset with Cut-off neighbourhood

For the Columbus CRIME variable, first centre the observations:

$$z_i = y_i - \bar{y}.$$

The first part of the centred vector in the original analysis is:

```
columbus$CRIME - mean(columbus$CRIME)
```

```
[1] -19.402844 -16.327070 -4.502043 -2.741064 15.602686 -9.062166
[7] -34.950555  3.297034 -4.612907 -1.127989 27.146624 21.576845
[13] 11.587305 21.937308 13.456663 19.709887  1.739950  8.833662
[19] 19.393141 -34.905027  4.945250 -1.423776 -15.080320  3.169047
[25] 26.170351  5.840918 17.665606 21.790961 25.621622 33.763220
[31] -17.451610 -15.983232  6.839339 -11.154796  4.046229 -20.823268
[37]  7.316252 18.582114 -16.027961 -18.887525 -16.223678 -18.636934
[43]  1.534788 -9.166561 -6.100336 -18.598291 -7.305963 -8.483558
[49] -12.587333
```

The critical-cut-off neighbourhood is then represented through a row-standardised spatial weights matrix $W^{(s)}$.

```
weightsmatrix_s = listw2mat(dnbTresh1.listw)
head(weightsmatrix_s)
```

```
      1  2  3      4      5 6  7  8 9 10 11 12 13 14 15 16
1 0.0000000 0.50 0.50 0.0000000 0.0000000 0 0.00 0.00 0 0 0.0 0.0 0 0 0.0 0
2 0.5000000 0.00 0.00 0.5000000 0.0000000 0 0.00 0.00 0 0 0.0 0.0 0 0 0.0 0
3 0.3333333 0.00 0.00 0.3333333 0.3333333 0 0.00 0.00 0 0 0.0 0.0 0 0 0.0 0
4 0.0000000 0.25 0.25 0.0000000 0.0000000 0 0.25 0.25 0 0 0.0 0.0 0 0 0.0 0
5 0.0000000 0.00 0.20 0.0000000 0.0000000 0 0.00 0.20 0 0 0.2 0.2 0 0 0.2 0
6 0.0000000 0.00 0.00 0.0000000 0.0000000 0 0.00 0.00 1 0 0.0 0.0 0 0 0.0 0
 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42
1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
3 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
4 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
5 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
6 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 43 44 45 46 47 48 49
```

```

1 0 0 0 0 0 0 0
2 0 0 0 0 0 0 0
3 0 0 0 0 0 0 0
4 0 0 0 0 0 0 0
5 0 0 0 0 0 0 0
6 0 0 0 0 0 0 0

```

Calculation of Moran's I for the Columbus Dataset with Cut-off neighbourhood

Using matrix notation with centred observations z and a row-standardised matrix $W^{(s)}$,

$$I = \frac{Z'W^sZ}{Z'Z}.$$

```

Y_s = columbus$CRIME - mean(columbus$CRIME)
lagY_s <- lag.listw(dnbTresh1.listw, Y_s)
moransI = t(Y_s)%*%weightsmatrix_s%*%Y_s/(t(Y_s)%*%Y_s)
moransI

```

```

      [,1]
[1,] 0.5518257

```

For the Columbus CRIME example in the lecture,

$$I = 0.5518257.$$

The reported randomisation test gives

$$E(I) = -0.02083333, \quad \text{Var}(I) = 0.01038068,$$

with a standard deviate of 5.6206 and a very small p -value.

```

moran.test(columbus$CRIME,dnbTresh1.listw)

```

Moran I test under randomisation

```

data: columbus$CRIME
weights: dnbTresh1.listw

```

```

Moran I statistic standard deviate = 5.6206, p-value = 9.514e-09
alternative hypothesis: greater
sample estimates:
Moran I statistic      Expectation      Variance
      0.55182569      -0.02083333      0.01038068

```

The compact matrix expression is useful because it shows Moran's I as a normalised spatial cross-product. It is also the bridge to spatial regression models, where Wy becomes the spatially lagged response.

We need to do matrix calculations in order to calculate the Moran's I statistic.

Moran Scatter Plot: Y_s vs $\text{lag}Y_s$

The Moran scatter plot places the centred or standardised variable on the horizontal axis and its spatial lag on the vertical axis:

$$Z_i \text{ versus } (WZ)_i.$$

When W and z are standardised consistently, the slope of the fitted line is Moran's I .

The plot therefore links the global statistic to the local behaviour of individual observations.

```

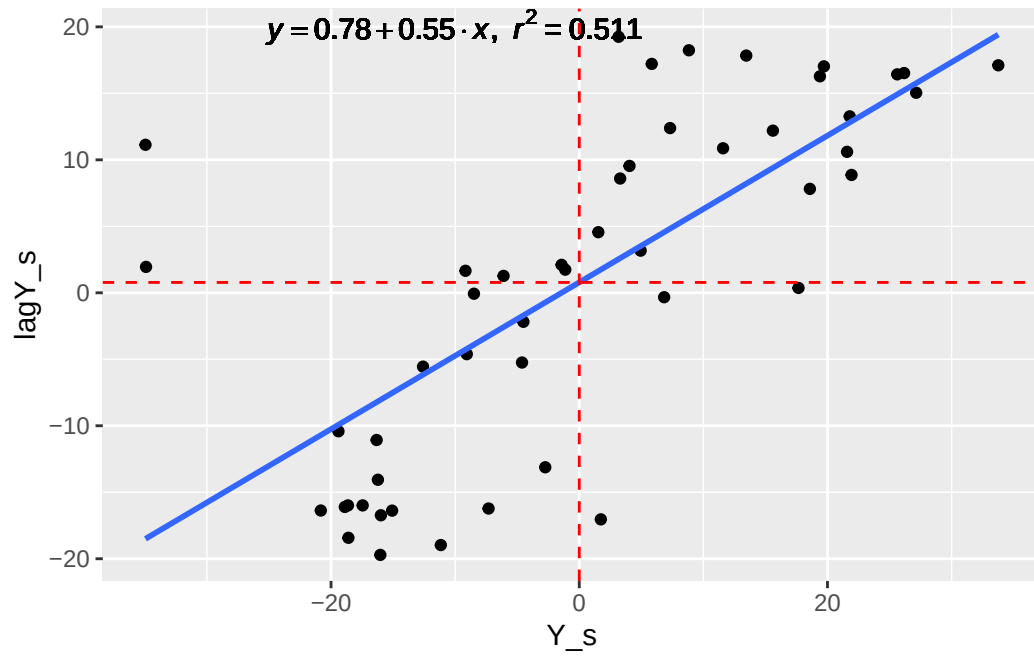
lm_eqn <- function(df, y, x){
  m <- lm(y ~ x, df);
  eq <- substitute(italic(y) == a + b %.% italic(x)*", "~italic(r)^2~"="~r2,
    list(a = format(unname(coef(m)[1]), digits = 2),
          b = format(unname(coef(m)[2]), digits = 2),
          r2 = format(summary(m)$r.squared, digits = 3)))
  as.character(as.expression(eq));
}

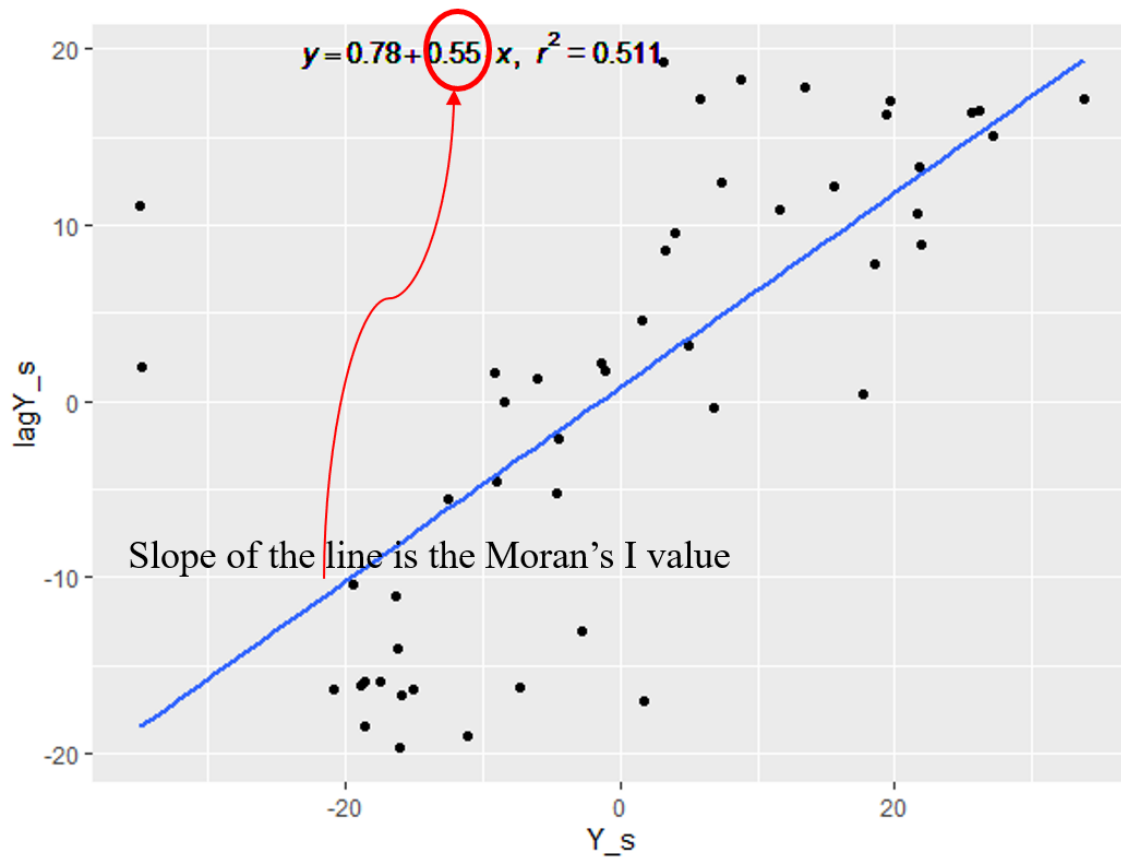
dataMoransI = data.frame(Y_s, lagY_s)

library(ggplot2)

ggplot(dataMoransI, aes(x = Y_s, y = lagY_s)) + geom_point() + geom_smooth(method = "lm", se

```

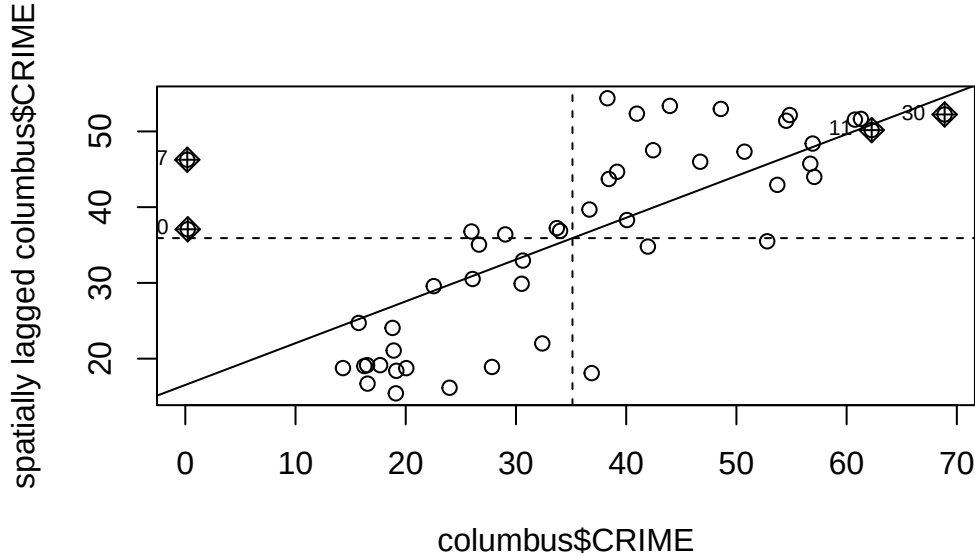




The Moran scatter plot is descriptive. Influential observations can strongly affect the fitted slope, so ordinary regression t -tests on that plotted line are not a substitute for a spatial randomisation test.

Moran's I Plot with `moran.plot ()` function in R

```
moran.plot(
  columbus$CRIME,
  dnbTresh1.listw,
  labels = row.names(columbus)
)
```

As you can see Moran's I is the slope of the linear association between the attribute and its lagged values based on a weights matrix, should you use significance tests on this slope? NO. Why NOT? Need to be careful with outliers, since we all know the outliers influence the slope. What this plot helps us do is to move from global to local thinking.

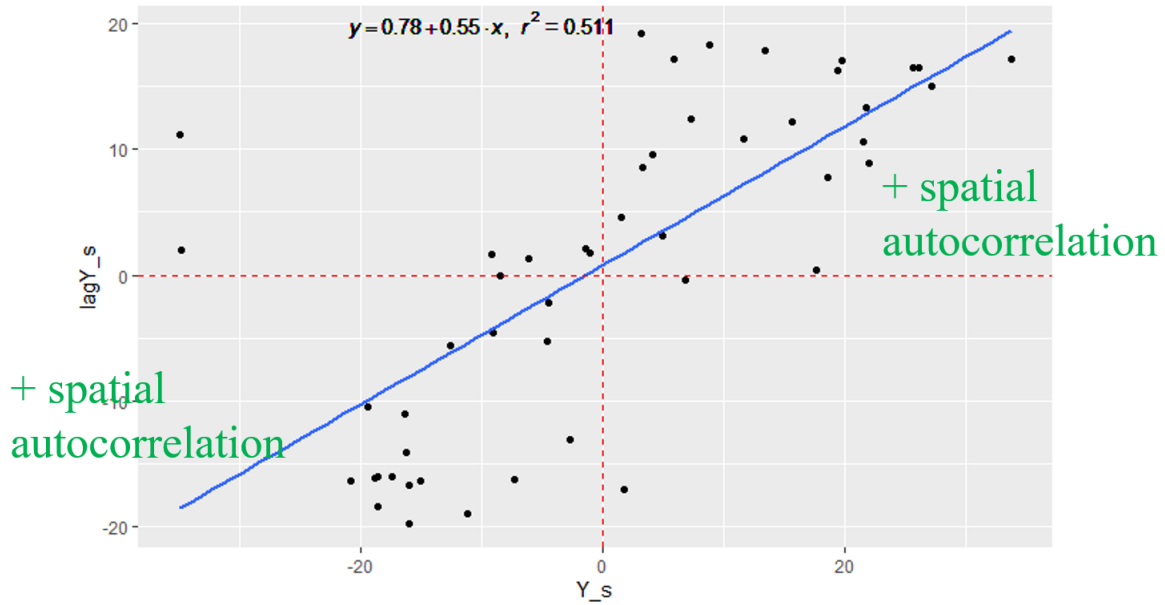
we can also apply standard techniques for detecting observations with unusually strong influence on the slope. Specifically, `moran.plot` calls `influence.measures` on the linear model of `lm(wy ~ y)` providing the slope coefficient, where `wy` is the spatially lagged value of `y`.

Moran's I

The Moran scatter plot is partitioned into four quadrants:

Quadrant	Observation	Neighbours	Interpretation
High-High	High	High	positive association / high-value cluster
Low-Low	Low	Low	positive association / low-value cluster
High-Low	High	Low	negative association / spatial outlier
Low-High	Low	High	negative association / spatial outlier

The High-High and Low-Low quadrants represent **positive spatial autocorrelation**. The High-Low and Low-High quadrants represent **negative spatial autocorrelation**.



Quadrants describe the sign pattern of a location and its spatial lag. Statistical significance is a separate question. A High–High point is not automatically a significant hot spot until a local inferential procedure is applied.

The plot is further partitioned into quadrants at the mean values of the variable and its lagged values: low–low, low–high, high–low, and high–high. We have the 4 quadrats of the scatter plot, where upper right and lower left indicate positive spatial autocorrelation indicating that locations are similar to their neighbours, whereas lower right and upper left show negative spatial autocorrelation which indicates the spatial outliers, locations are different from their neighbours.

Geary's C

Geary's C focuses on **squared differences between neighbouring observations**:

$$C = \frac{(n-1)}{2S_0} \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (y_i - y_j)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}.$$

where

$$S_0 = \sum_i \sum_j w_{ij}.$$

Interpretation is opposite in direction to Moran's I :

- $C < 1$: positive spatial autocorrelation;
- $C \approx 1$: spatial randomness;
- $C > 1$: negative spatial autocorrelation.

Geary's C reacts strongly to local neighbour-to-neighbour differences. Its null reference value is approximately 1, which is why its interpretation runs in the opposite numerical direction to Moran's I .

```
geary.test(
  columbus$CRIME,
  dnbTresh1.listw,
  alternative = "less"
)
```

Geary C test under randomisation

```
data: columbus$CRIME
weights: dnbTresh1.listw
```

```
Geary C statistic standard deviate = 5.2484, p-value = 1
alternative hypothesis: Expectation less than statistic
sample estimates:
```

Geary C statistic	Expectation	Variance
0.44927847	1.00000000	0.01101077

Spatial Correlogram

A non-parametric spatial correlogram is an alternative way of studying global spatial dependence.

Rather than beginning with one predefined weights matrix, it examines the association between pairs of observations as a function of their separation distance.

Conceptually,

$$\text{Cov}(z_i, z_j) = f(d_{ij}) + \varepsilon_{ij},$$

where

- d_{ij} is the pairwise distance between locations i and j ;
- $f(\cdot)$ is a non-parametric function estimated from the data;
- a LOWESS or kernel smoother may be used.

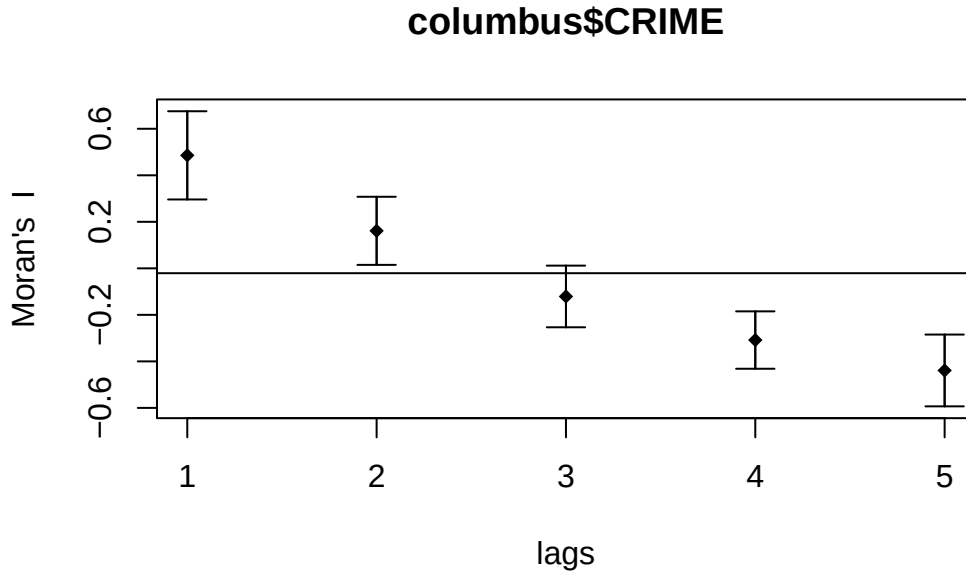
This allows us to ask how spatial association changes as distance increases.

A distance-based correlogram helps reveal the *scale* of spatial dependence. Positive association at short distances may decay, disappear, or even become negative as distance increases.

Both Moran's I and Geary's C depend on weights matrices, coming from neighbourhood relationships. Most people would find this description arbitrary. In fact there are tons of different neighbourhood descriptions one can make. It might be worthwhile investigating other ways of measuring spatial autocorrelation.

Spatial Correlogram

```
spcorrelogram = sp.correlogram(  
  col.gal.nb,  
  columbus$CRIME,  
  order = 5,  
  method = "I",  
  style = "W"  
)  
  
plot(spcorrelogram)
```



Here we see the values of Moran's I for ten successive lag orders of contiguous neighbours. The Figure shows the output plot from `spdep` package with `sp.correlogram()` function, and suggests that second-order neighbours are also positively autocorrelated. The horizontal line is the expected Moran's I under no spatial autocorrelation.

Local Measures and Tests for Spatial Autocorrelation

Local Indicators of Spatial Association (**LISA**) were introduced by Anselin (1995) to decompose global spatial autocorrelation into contributions from individual observations.

A common form of the local Moran statistic for area i is

$$I_i = \frac{z_i}{m_2} \sum_j w_{ij} z_j,$$

where

$$z_i = y_i - \bar{y}, \quad m_2 = \frac{1}{n} \sum_i z_i^2.$$

The summation runs over the neighbourhood of area i .

The collection of local Moran statistics is related to the global Moran statistic: their sum is proportional to global Moran's I .

Global Moran's I can be significant while hiding substantial local heterogeneity. LISA statistics help reveal where the global pattern is being generated and where spatial outliers occur.

With the advent of large data sets characteristic of GISystems, it has become clear that the need to assess spatial autocorrelation globally may be of only marginal interest. During the past two decades, a number of statistics, called local statistics, have been developed. These provide for each observation of a variable an indication of the extent of significant spatial clustering of similar values around that observation. Hence, they are well suited to identify the existence of hot spots (local clusters of high values) or cold spots (local clusters of low values), and are appropriate to identify distances beyond which no discernible association exists. Local indicators of spatial association (LISA) statistics were derived by Anselin (1995), in his paper of Local Indicators of Spatial Association—LISA published in *Geographical Analysis* in 1995. An initial evaluation of the properties of a LISA statistic is carried out for the local Moran, which is applied in a study of the spatial pattern of conflict for African countries and in a number of Monte Carlo simulations. HE suggests that a local indicator of spatial association (LISA) is any statistic that satisfies the following two requirements a) the LISA for each observation gives an indication of the extent of significant spatial clustering of similar values around that observation; b. the sum of LISAs for all observations is proportional to a global indicator of spatial association. The local Moran statistic is defined as follows: where N_i denotes the neighbourhood set of area i (can be formalized by means of a spatial weights or contiguity matrix), $\sum_{j \in N_i} x_j$ denotes the summation in runs only over those areas belonging to N_i , $\bar{x}_i$ denotes the average of these neighbouring observations. The LISA statistic, I_i serve two purposes. On the one hand, they may be viewed as indicators of local pockets of non-stationarity, or hot spots. On the other hand, they may be used to assess the influence of individual locations (observations) on the magnitude of the corresponding global spatial autocorrelation statistic, Moran's I . With the advent of large data sets characteristic of GISystems, it has become clear that the need to assess spatial autocorrelation globally may be of only marginal interest. During the past two decades, a number of statistics, called local statistics, have been developed. These provide for each observation of a variable an indication of the extent of significant spatial clustering of similar values around that observation. Hence, they are well suited to identify the existence of hot spots (local clusters of high values) or cold spots (local clusters of low values), and are appropriate to identify distances beyond which no discernible association exists. Local indicators of spatial association (LISA) statistics were derived by Anselin (1995), in his paper of Local Indicators of Spatial Association—LISA published in *Geographical Analysis* in 1995. An initial evaluation of the properties of a LISA statistic is carried out for the local Moran, which is applied in a study of the spatial pattern of conflict for African countries and in a number of Monte Carlo simulations. HE suggests that a local indicator of spatial association (LISA) is any statistic that satisfies the following two requirements a) the LISA for each observation gives an indication of the extent of significant spatial clustering of similar values around that observation; b. the sum of LISAs for all observations is proportional to a global indicator of spatial association. The local Moran statistic is defined as follows: where N_i denotes the

neighbourhood set of area a_i (can be formalized by means of a spatial weights or contiguity matrix) , and the summation in $\sum$ runs only over those areas belonging to a_i , $\bar{y}_i$ denotes the average of these neighbouring observations. The LISA statistic, I_i serve two purposes. On the one hand, they may be viewed as indicators of local pockets of non-stationarity, or hot spots. On the other hand, they may be used to assess the influence of individual locations (observations) on the magnitude of the corresponding global spatial autocorrelation statistic, Moran's I

Identification of Local Spatial Clusters

Local spatial clusters—often called **hot spots** or **cold spots**—may be identified as locations, or sets of contiguous locations, for which a LISA statistic is significant.

The null hypothesis is **no local spatial association**.

Because the exact distribution of a generic LISA statistic can be difficult to obtain, a common alternative is a **conditional randomisation / permutation procedure** that produces empirical pseudo- p -values.

In conditional permutation for local Moran's I_i , the focal value at location i is held fixed while values at other locations are permuted. The observed I_i is then compared with this empirical reference distribution.

```
# Conditional permutation inference for local Moran statistics
local_perm <- localmoran_perm(
  columbus$CRIME,
  dnbTresh1.listw,
  nsim = 999
)

head(local_perm)
```

	Ii	E.Ii	Var.Ii	Z.Ii	Pr(z != E(Ii))	
1	0.73681849	-0.0349709214	0.67858572	0.9369069	0.34880642	
2	0.65915397	-0.0229447477	0.48805819	0.9763633	0.32888448	
3	0.03579329	0.0001301186	0.02388031	0.2307809	0.81748499	
4	0.13113818	-0.0004941870	0.00677204	1.5995688	0.10969428	
5	0.69380337	-0.0254049290	0.17771437	1.7060565	0.08799755	
6	0.15242670	-0.0474488184	0.29400244	0.3686244	0.71240767	
	Pr(z != E(Ii))	Sim	Pr(folded)	Sim	Skewness	Kurtosis
1		0.370		0.185	-0.02407492	-0.37667127
2		0.332		0.166	0.10819510	-0.10504981
3		0.828		0.414	-0.03773995	-0.26497239

4	0.118	0.059	-0.09411295	-0.15231936
5	0.082	0.041	-0.03513799	-0.08864078
6	0.734	0.379	0.09398989	-0.75588569

A test for significant local spatial association may be based on the moments, although the exact distribution of such a statistic is still unknown (Anselin 1995, p. 99). Alternatively, a conditional random permutation test can be used to yield so-called pseudo significance levels (for example, as in Hubert 1987). The randomization is conditional in the sense that the value Y_i at a location i is held fixed (that is, not used in the permutation) and the remaining values are randomly permuted over the locations in the data set. For each of these resampled data sets, the value of L_i can be computed. The resulting empirical distribution function provides the basis for a statement about the extremeness (or lack of extremeness) of the observed statistic, relative to (and conditional on) the values computed under the null hypothesis (the randomly permuted values). In practice, this is straightforward to implement, since for each location only as many values as there are in the neighborhood set need to be resampled. Note that this same approach can also easily be applied to the G_i and G_T statistics.

Local Moran's I

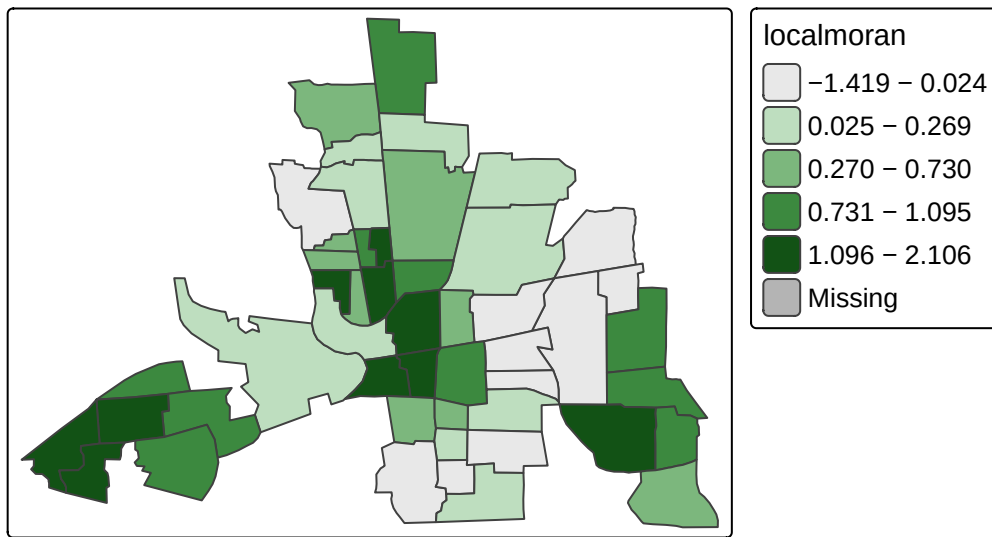
```
localmoranstatistics = localmoran(Y_s,dnbTresh1.listw)
head(localmoranstatistics)
```

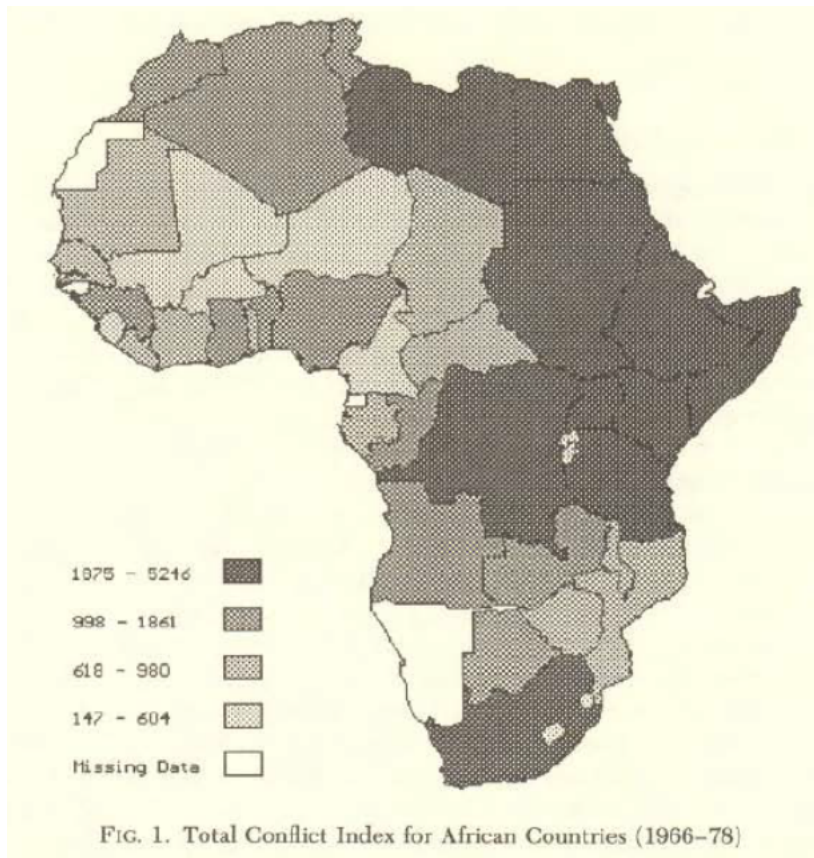
	Ii	E.Ii	Var.Ii	Z.Ii	Pr(z != E(Ii))
1	0.73681849	-0.0285985420	0.666144891	0.9378077	0.34834327
2	0.65915397	-0.0202502140	0.475741297	0.9850149	0.32461676
3	0.03579329	-0.0015396867	0.024041007	0.2407777	0.80972741
4	0.13113818	-0.0005707572	0.006541757	1.6284260	0.10343458
5	0.69380337	-0.0184931910	0.162742823	1.7656716	0.07745096
6	0.15242670	-0.0062384562	0.303777355	0.2878749	0.77344247

```
columbus$localmoran = localmoranstatistics[,1]
```

Plot of the Local Moran Statistics

```
tm_shape(columbus) + tm_polygons(style="quantile", col = "localmoran") +
  tm_legend(outside = TRUE, text.size = .8)
```



Measures of spatial association, such as Moran's I, have been applied to quantitative indices for various types of conflicts and cooperation between nation-states in Africa, such as those contained in the COPDAB data base (Azar 1980). The spatial pattern of the index for total conflict is illustrated in the quartile map in Figure 1, with the darkest shade corresponding to the highest quartile [for details on the data sources, see Anselin and O'Loughlin (1992)]. The suggestion of spatial clustering of similar values that follows from a visual inspection of this map is confirmed by a strong positive and significant Moran's I of 0.417, with an associated standard normal z-value of 4.35 ($p < 0.001$). These statistics are computed for a row-standardized spatial weights matrix based on first-order contiguity (common border), given the importance of borders in the study of international conflict. All computations were carried out with the SpaceStat software for spatial data analysis.

TABLE 1
Measures of Local Spatial Association

Id	Country	G_i^*	p	I_i	$z(I_i)$	p_n	p_r
1	Gambia	-0.984	0.1626	0.375	0.428	0.3342	0.4727
2	Mali	-1.699	0.0447	0.464	1.482	0.0692	0.0456
3	Senegal	-1.463	0.0717	0.257	0.623	0.2667	0.0270
4	Benin	-1.301	0.0966	0.194	0.484	0.3142	0.0612
5	Mauritania	-0.605	0.2726	0.097	0.269	0.3940	0.4111
6	Niger	-1.049	0.1471	0.231	0.774	0.2193	0.2404
7	Ivory Coast	-1.417	0.0782	0.290	0.788	0.2154	0.0611
8	Guinea	-1.449	0.0737	0.183	0.519	0.3020	0.0365
9	Burkina Faso	-1.751	0.0400	0.508	1.479	0.0695	0.0339
10	Liberia	-1.041	0.1490	0.186	0.398	0.3452	0.1333
11	Sierra Leone	-0.870	0.1921	0.265	0.444	0.3286	0.4006
12	Ghana	-1.103	0.1351	0.148	0.326	0.3721	0.0885
13	Togo	-0.991	0.1610	0.219	0.462	0.3219	0.1894
14	Cameroon	-1.133	0.1285	0.259	0.711	0.2387	0.1706
15	Nigeria	-1.173	0.1205	0.114	0.306	0.3798	0.0851
16	Gabon	-0.789	0.2150	0.204	0.349	0.3634	0.3139
17	CAR	1.174	0.1203	-0.442	-1.046	0.1477	0.0613
18	Chad	0.463	0.3218	-0.105	-0.225	0.4111	0.2125
19	Congo	-0.203	0.4198	0.011	0.079	0.4684	0.4734
20	Zaire	2.023	0.0216	0.710	2.591	0.0048	0.0404
21	Angola	1.235	0.1085	0.118	0.270	0.3936	0.0999
22	Uganda	3.336	0.0004	1.943	4.928	0.0000	0.0031
23	Kenya	3.503	0.0002	1.197	3.060	0.0011	0.0016
24	Tanzania	1.098	0.1360	0.272	0.973	0.1652	0.1898

Using the same row-standardized weights matrix as for the global measures given earlier, the results for the indicators of local spatial association are reported in the third and fifth columns of Table 1, for each of the forty-two countries in the example. The standardized z-value for I_i , computed by subtracting the expected value (13) and dividing by the standard deviation [the square root of (14)], is listed in the sixth column. Two indications of significance are given, one based on an approximation by the normal distribution, P_n (in the seventh column of Table 1) and one derived from conditional randomization, using a sample of 10,000 permutations, P_r (in the last column of Table 1). Given this conservative procedure, the normal approximation for both the G_i and the I_i show the same four countries to exhibit local spatial clustering (with the significance levels in bold type in Table 1). They are Uganda (22), Kenya (23), Sudan (41), and Egypt (42), which themselves form a cluster in the northeast of Africa, part of the so-called Shatterbelt. Shatter belt is a concept in geopolitics according to which on the political map are recognized and analyzed strategically positioned and oriented regions that are deeply internally divided and encompassed in the competition between the great powers in the geostrategic areas and spheres

Local Moran in R

In R, local Moran statistics can be calculated with `spdep::localmoran()`:

```
localmoranststatistics <- spdep::localmoran(
  Y_s,
  dnbTresh1.listw
)
```

```
localmoranststatistics
```

	Ii	E.Ii	Var.Ii	Z.Ii	Pr(z != E(Ii))
1	0.736818491	-2.859854e-02	0.6661448908	0.93780765	0.3483432685
2	0.659153968	-2.025021e-02	0.4757412970	0.98501488	0.3246167634
3	0.035793288	-1.539687e-03	0.0240410072	0.24077770	0.8097274126
4	0.131138176	-5.707572e-04	0.0065417568	1.62842603	0.1034345818
5	0.693803371	-1.849319e-02	0.1627428232	1.76567158	0.0774509642
6	0.152426700	-6.238456e-03	0.3037773549	0.28787494	0.7734424674
7	-1.418605235	-9.279429e-02	0.7547857820	-1.52605345	0.1269965546
8	0.103319296	-8.257717e-04	0.0050383143	1.46722284	0.1423154446
9	0.088187389	-1.616451e-03	0.0386977724	0.45651171	0.6480220495
10	-0.007156422	-9.665465e-05	0.0047356202	-0.10258928	0.9182889508
11	1.489116910	-5.598153e-02	0.2387512105	3.16215652	0.0015660536
12	0.834073870	-3.536625e-02	0.1351553596	2.36495586	0.0180322198
13	0.459183376	-1.019948e-02	0.0526252173	2.04611609	0.0407449499
14	0.708918984	-3.655778e-02	0.1591203691	1.86883613	0.0616456170
15	0.875366588	-1.375586e-02	0.0828431006	3.08911022	0.0020075694
16	1.223927852	-2.951083e-02	0.1293875235	3.48463177	0.0004928147
17	-0.108094629	-2.299782e-04	0.0035956407	-1.79883361	0.0720450046
18	0.587470453	-5.927816e-03	0.0266215244	3.63688706	0.0002759529
19	1.151223098	-2.856995e-02	0.1694747499	2.86584918	0.0041589234
20	-0.247951607	-9.255269e-02	0.7530211176	-0.17907889	0.8578757561
21	0.057144088	-1.857760e-03	0.0908611332	0.19573863	0.8448147484
22	-0.010918574	-1.539914e-04	0.0011236363	-0.32113253	0.7481099581
23	0.900849470	-1.727564e-02	0.2654944353	1.78186236	0.0747716792
24	0.222445111	-7.629051e-04	0.0046550365	3.27150994	0.0010697483
25	1.576737186	-5.202741e-02	0.2570965315	3.21225647	0.0013169675
26	0.366611808	-2.591644e-03	0.0188644701	2.68808779	0.0071862501
27	0.023496701	-2.370665e-02	0.1689066665	0.11485488	0.9085601515
28	1.054294593	-3.607164e-02	0.1219320577	3.12257905	0.0017927395
29	1.534813119	-4.986851e-02	0.2893304273	2.94608293	0.0032182610
30	2.106237710	-8.659660e-02	0.5772443881	2.88619691	0.0038992811
31	1.017447516	-2.313578e-02	0.5419314656	1.41352881	0.1575002960
32	0.975052651	-1.940628e-02	0.2975917603	1.82295690	0.0683099184
33	-0.008281905	-3.553378e-03	0.0258399512	-0.02941576	0.9765330041
34	0.771608734	-9.452275e-03	0.1464202875	2.04119422	0.0412315257

```

35  0.140766915 -1.243695e-03 0.0090650426  1.49154332  0.1358189064
36  1.243242522 -3.293904e-02 0.2856021104  2.38798590  0.0169409917
37  0.330451865 -4.066216e-03 0.0247289081  2.12724311  0.0333998888
38  0.529422011 -2.623031e-02 0.1153931242  1.63573582  0.1018948896
39  1.151957408 -1.951505e-02 0.4588139896  1.72947288  0.0837244959
40  1.109043195 -2.709962e-02 0.6322045881  1.42890746  0.1530308318
41  0.831677994 -1.999456e-02 0.3064288938  1.53853646  0.1239175009
42  1.086553172 -2.638530e-02 0.6159922141  1.41802388  0.1561837935
43  0.025506418 -1.789410e-04 0.0008082627  0.90346090  0.3662813328
44 -0.055325820 -6.383017e-03 0.0462851033 -0.22749310  0.8200403375
45 -0.028304126 -2.826966e-03 0.0252748048 -0.16025335  0.8726815086
46  1.249537703 -2.627600e-02 0.6135092567  1.62883353  0.1033482644
47  0.432196383 -4.054787e-03 0.1978789359  0.98070146  0.3267399841
48  0.002170906 -5.467253e-03 0.0396811746  0.03834391  0.9694134795
49  0.254910325 -1.203595e-02 0.1363680392  0.72288227  0.4697522160

```

```
attr(,"call")
```

```
spdep::localmoran(x = Y_s, listw = dnbTresh1.listw)
```

```
attr(,"class")
```

```
[1] "localmoran" "matrix"      "array"
```

```
attr(,"quadr")
```

	mean	median	pysal
1	Low-Low	Low-Low	Low-Low
2	Low-Low	Low-Low	Low-Low
3	Low-Low	Low-Low	Low-Low
4	Low-Low	Low-Low	Low-Low
5	High-High	High-High	High-High
6	Low-Low	Low-Low	Low-Low
7	Low-High	Low-High	Low-High
8	High-High	High-High	High-High
9	Low-Low	Low-Low	Low-Low
10	Low-High	Low-Low	Low-High
11	High-High	High-High	High-High
12	High-High	High-High	High-High
13	High-High	High-High	High-High
14	High-High	High-High	High-High
15	High-High	High-High	High-High
16	High-High	High-High	High-High
17	High-Low	High-Low	High-Low
18	High-High	High-High	High-High
19	High-High	High-High	High-High
20	Low-High	Low-High	Low-High
21	High-High	High-High	High-High
22	Low-High	Low-High	Low-High

```

23  Low-Low  Low-Low  Low-Low
24  High-High High-High High-High
25  High-High High-High High-High
26  High-High High-High High-High
27  High-Low  High-Low High-High
28  High-High High-High High-High
29  High-High High-High High-High
30  High-High High-High High-High
31  Low-Low  Low-Low  Low-Low
32  Low-Low  Low-Low  Low-Low
33  High-Low  High-Low High-Low
34  Low-Low  Low-Low  Low-Low
35  High-High High-High High-High
36  Low-Low  Low-Low  Low-Low
37  High-High High-High High-High
38  High-High High-High High-High
39  Low-Low  Low-Low  Low-Low
40  Low-Low  Low-Low  Low-Low
41  Low-Low  Low-Low  Low-Low
42  Low-Low  Low-Low  Low-Low
43  High-High High-High High-High
44  Low-High  Low-Low  Low-High
45  Low-High  Low-Low  Low-High
46  Low-Low  Low-Low  Low-Low
47  Low-Low  Low-Low  Low-Low
48  Low-Low  Low-Low  Low-Low
49  Low-Low  Low-Low  Low-Low

```

The output provides a local statistic for every spatial unit, together with quantities used for inference.

Modern `spdep::localmoran()` returns several columns rather than only I_i . When teaching, it is useful to separate the local statistic itself, its expectation/variance under the chosen framework, and the resulting inferential quantities.

```

localmoranstatistics <- localmoran(
  Y_s,
  dnbTresh1.listw
)

```

```
head(localmoranstatistics)
```

Ii	E.Ii	Var.Ii	Z.Ii	Pr(z != E(Ii))
----	------	--------	------	----------------

1	0.73681849	-0.0285985420	0.666144891	0.9378077	0.34834327
2	0.65915397	-0.0202502140	0.475741297	0.9850149	0.32461676
3	0.03579329	-0.0015396867	0.024041007	0.2407777	0.80972741
4	0.13113818	-0.0005707572	0.006541757	1.6284260	0.10343458
5	0.69380337	-0.0184931910	0.162742823	1.7656716	0.07745096
6	0.15242670	-0.0062384562	0.303777355	0.2878749	0.77344247